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Short communication

Antidoping testing and testicle cancer diagnosis: An unexpected helper

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ABSTRACT

Human chorionic gonadotrophin (hCG) is prohibited in male athletes due to its potential performance-enhancing effects. Some athletes with elevated hCG levels during routine doping tests have been diagnosed with testicular cancer, highlighting the inadvertent role of anti-doping screening in cancer detection. This study explores the incidental detection of testicular cancer through routine anti-doping tests among elite athletes, aiming to raise awareness about the potential secondary benefits of these screenings. We have analyzed cases from the past decade to demonstrate how early detection has facilitated timely medical interventions, allowing athletes to heal and return to competitive sports. Through these instances, athletes become unintentional advocates, contributing to public health awareness about this topic.
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Antidoping testing plays a crucial role in modern sports, striving to ensure a level playing field by detecting and preventing the use of performance-enhancing drugs. Athletes are subject to rigorous testing protocols, including urine and blood samples to identify substances that could give them an unfair advantage. These tests are intended to maintain fairness in competition and protect athletes' health by discouraging the use of substances that can have harmful side effects.[1]

Human chorionic gonadotrophin (hCG) is a glycoprotein that stimulates the production of testosterone by Leydig cells in males, which are crucial for developing and maintaining male reproductive health. Pharmaceutical preparations of hCG can stimulate testosterone production without increasing the testosterone-to-epitestosterone ratio and prevent testicular shutdown and atrophy during and after prolonged courses of androgen administration.[2] A positive effect of hCG on muscle strength has not been demonstrated in females and elevated hCG concentrations are normally caused by pregnancy. Therefore, the use of hCG is prohibited only in male athletes. Normally, hCG is either undetectable or found at very low levels in urine from healthy, non-doping males. Unfortunately, besides substance abuse, elevated levels of hCG and hCG-core fragments can be produced by testicular cancers and rarely from extra-testicular germ cell tumors.[3] Indeed, in recent years there have been some media reports of top athletes being diagnosed with testicular cancer as a result of testing positive for high hCG concentrations detected during routine doping controls.

Our short communication aimed to explore specifically this topic. A systematic web-based search via Google "Advanced Search" function with specific keywords has been conducted, as reported in Supplementary Table 1. Only cases in which there was a clear connection between a positive hCG doping test result in a male athlete and a diagnosis of testicular cancer were included. The exclusion criterion was any case of testicular cancer diagnosis in athletes that did not directly result from a positive doping control test. In Table 1 and Supplementary Table 2, we reported the public-available identified athletes who were diagnosed with testicular cancer following a positive doping test since 2010. All these athletes were under 35 and almost all were diagnosed at an early stage, allowing a complete and rapid return to play in all cases. Only one case resulted in a diagnosis of metastatic disease that led to the premature death of the young athlete.[4] Moreover, the case of Francesco Acerbi allowed for the diagnosis of a recurrence and prompt treatment, with the application of hCG as a biomarker in follow-up evaluations.

In September 2011 World Anti-Doping Agency (WADA) released guidelines for the reporting and management of elevated urinary concentrations of hCG in male athletes (updated to the 2021 technical document and extended to luteinizing hormone from 2015).[5] Indeed, following an adverse analytical finding (AAF), it is necessary to promptly exclude a pathological condition. The athlete should be immediately alerted and undergo a specialist medical evaluation for the assessment of history and physical examination followed by new laboratory tests and imaging techniques.

Abbreviations: hCG, human chorionic gonadotrophin; WADA, World Anti-Doping Agency; AAFs, adverse analytical findings.

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Table 1
Diagnosis of testicular cancer following a positive doping test from 2010 to date.

Athlete	Date of positive doping test	Nationality	Age	Discipline	Details
#1 Emmanuel Garcia	June 2010	Canada	24	Baseball	Personal history of cryptorchidism and previous surgery for testicular descent. Diagnosis of early-stage seminoma. He underwent surgery in June 2010 and daily radiotherapy treatments for 30 days. He returned to game action in the next winter.
#2 Jake Gibb	December 2010	USA	34	Beach Volley	He underwent surgery. Previous malignant melanoma removed. He participated in the 2012 Olympic Games.
#3 Magrão Márcio Rodrigues	2011	Brazil	33	Football	Cancer was surgically removed and publicly revealed four years later after a new positive test result possibly caused by drug therapy.
#4 Dylan Tombides	June 2011	Australia	17	Football	Diagnosis of metastatic cancer. He underwent cycles of chemotherapy, stem cell transplants, radiotherapy, vascular and major abdominal surgery. In September 2012 he made his debut with West Ham. In March 2013 new liver metastasis was diagnosed. The worsening of his condition caused his death in April 2014. In February 2015, a foundation has been set up in his name with the aims of fundraising and spreading awareness of testicular cancer.
#5 Francesco Acerbi	December 2013	Italy	25	Football	Recurrence in the right testicle of left testicle cancer. After three months of chemotherapy, he returned to the field at a high level.
#6 Thomas Van Der Plaetsen	October 2014	Belgium	24	Athletics (Decathlon)	He underwent operation and chemotherapy. He took part in the World Championships in Beijing and he won the decathlon gold medal at the 2016 European Championships.
#7 Marco Russ	May 2016	Germany	30	Football	He underwent surgery promptly and two rounds of chemotherapy. He returned to competitive activity in December 2016.
#8 Ederson Honorato Campos	July 2017	Brazil	31	Football	He underwent 21 cycles of chemotherapy and two surgical operations. He returned to competitive activity in January 2018.
#9 Maarten Stuivenberg	August 2017	Netherlands	23	Athletics (400 m runner)	Six days of hospitalization for chemotherapy. He returned to competitive activity in March 2019.
#10 Shea Theodore	May 2019	Canada	24	Ice-Hockey	He immediately underwent surgery to remove the tumor (mixed germ cell tumor, stage I) and he had a full recovery for the start of the following season.
#11 Yannick Kaeser	May 2021	Switzerland	28	Swimming	After an early diagnosis, he decided to participate European Championships before undergoing surgery. He returned to competitive activity in November 2021.
#12 Torstein Træen	May 2022	Norway	27	Cycling	15 mm tumor in one testicle surgically removed with no chemotherapy needed. He returned to competitive activity in August 2022.
#13 Achille Pollonara	October 2023	Italy	31	Basketball	Athlete underwent surgery and returned to competitive activity in December 2023.

Testicular cancer is one of the most common malignancies among young men (aged between 15 and 40), making it a concern for athletes who fall within this age group. However, its link with physical and sporting activities is still debated.[6] Testicular cancer is highly treatable, especially when detected at an early stage. Therefore, awareness, education, and early diagnosis are paramount in reducing its mortality rate: the disease, in its initial stages, often presents with subtle symptoms or may be entirely asymptomatic.[7] However, there is limited evidence to suggest that there may be benefits to encouraging testicular self-examination in the general healthy population as a screening procedure; therefore, the role of hCG testing is of particular interest in the early diagnosis of this tumor.[8]

Although anti-doping control tests are primarily optimized for detecting the use of prohibited substances in athletes, the incidental detection of health issues highlights a potential ancillary utility. Indeed, testicular cancer is not the only cancer detectable by a positive anti-doping test; anabolic agents, erythropoietin or growth hormone could be produced by secreting adrenal, kidney or pituitary gland tumors, resulting in a positive anti-doping result. Notwithstanding, testicular one is probably the most frequent cancer detected unintentionally through the positivity of the anti-doping testing. Considering that the total number of AAFs for hCG between 2014 and 2021 (first and latest available WADA Anti-Doping Testing Figures specifically reporting AAFs as different from atypical findings for hCG) was 100, and that there were at least six cases of diagnosed testicular cancer following a positive anti-doping result, this appears to be far more than a rare event.[9]

Despite this finding, this communication is not intended to propose the adoption of routine cancer screening in athletes, and cost-effectiveness considerations would only be speculative in that sense, since in WADA reports the incidence of anti-doping rule violations by substance is not made explicit. It is reasonable to assume that even a possible screening for male athletes would not be appropriate given the low overall hCG positivity (11 AAFs out of 241.430 samples analyzed in 2021).[9] Moreover, health screening tasks extend beyond WADA's traditional mandate. To address these concerns, it should be essential to provide robust justification for such an expansion as it could potentially affect its perceived legitimacy globally.[10]

Conclusion

The link between anti-doping testing and the diagnosis of testicular cancer underscores a secondary, yet significant, health benefit of these tests. Although these unintentional diagnoses are rare, they draw public attention to the broader issue, demonstrating that even healthy young people can face such challenges. Importantly, these individuals have the potential to fully recover and resume high-level sport activities if cancer is detected early and treated effectively. Athletes, serving as unwitting advocates for this cause, can play a pivotal role in raising awareness and promoting public health.[11] Our investigation into the dual role of anti-doping testing not only reinforces its primary purpose of promoting fair play but also highlights a potential avenue for early health screening. However, integrating health checks into anti-doping protocols should be approached with caution to ensure they enhance rather than complicate the primary objectives of these programs. Further research and collaboration are essential to explore these possibilities responsibly.

CRediT authorship contribution statement

Marco Vecchiato: conceptualization, methodology, original draft preparation, reviewing; Veronica Baioccato: conceptualization, data curation and reviewing and editing; Stefano Palermi: conceptualization, supervision and reviewing.

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Confirmation of ethical compliance

Not needed.

Declaration of interest statement

The authors have no competing interests to declare that are relevant to the content of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2024.08.001.

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